WawID

Installation guide — Senegal

Deploying the WawID platform and border posts in Senegal

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This guide describes how to set up WawID at Senegal's border posts: deployment phases, hosting, legal prerequisites, hardware, installation of counter workstations, eGate kiosks and the DPAF space, software installation of the platform, mobile apps, onboarding of airlines and the Treasury, training, acceptance testing and go-live. It is intended for the project teams of the DPAF, the operator and the integrators.

The high-availability hosting infrastructure (two active data centers in Senegal and a backup site), the backup strategy, failover procedures and detailed phasing are covered by the page *Deployment plan — high-availability infrastructure in Senegal*; this guide summarizes them. For other technical details already documented, it refers to the pages *Deployment and configuration*, *Security, personal data and compliance*, *eGate kiosk: setup and journey* and *API reference (airlines, devices, connectors)*.

Reading this guide

Each item is classified according to its actual state:

Label	Meaning
Existing	Works today in the demonstrator (code in the repository).
Recommendation	Advised practice for operations; it commits no party until it has been decided.
To be developed	Required for national production but absent from the current code.
To be checked	Legal, administrative or contractual point to confirm with the competent authority before any decision.

Starting point

The current WawID deployment is a **demonstrator** hosted on Netlify (Netlify Functions and Netlify Blobs). It validates the journeys (counter, eGate, e-Visa, eGate pre-enrollment, APIS/PNR, intelligence, security fee), not load, sovereignty or service continuity. National operations require the sovereign architecture described in `docs/ARCHITECTURE.md`.

Component	Today (Existing)	National target
Hosting	Netlify, outside the national territory	Two active data centers in Senegal (site A: national data center in Diamniadio; site B: Tier III-certified commercial data center) + backup site C (To be developed)
Backups	No dedicated backup	3-2-1-1-0 rule, immutable and offline copies at site C, restoration tested every quarter (To be developed)
Storage	One collection = one JSON document (Netlify Blobs)	Partitioned PostgreSQL + event bus (To be developed)
Officer identity	Username + password (scrypt), 12-hour session	OIDC, MFA, PKI officer cards (To be developed)

Component	Today (Existing)	National target
Biometrics	Simulated scores	Certified SDK / ABIS, liveness detection ISO/IEC 30107-3 (To be developed)
Chip reading	Local NFC bridge, simulation mode only	Real reader driver, CSCA verification via the ICAO PKD (To be developed)
External files	Connectors in "local copy" mode	Connection to the Interpol NCB, WAPIS, FPR, DAF (To be developed, agreements required)
Offline mode	None	Post server (edge) with a local copy of the lists (To be developed)
Payments	Simulated	Integration of payment operators (To be developed)
Demo data	Created automatically on the first API call if the database is empty	Demo dataset disabled in production (To be developed)
Inventory of posts and devices	Six posts defined in the code (`POSTES` in `public/shared/engine.js`); eGates and devices come from the demo dataset	Administrable register of posts, counters and devices (To be developed)
Security fee invoices	From the demo dataset; collections and provisional fee base work	Issuing real invoices from declarations and counts (To be developed)

> No real personal data may be entered into the demonstrator. The steps below that concern the demonstrator are for tests, demonstrations and training.

Deployment phases

The posts declared in WawID are those of the `POSTES` list:

Code	Post	Type	Region
`AIBD`	Blaise Diagne International Airport	Air	Thiès
`PDK`	Autonomous Port of Dakar	Sea	Dakar
`ROS`	Rosso	Land	Saint-Louis
`KID`	Kidira	Land	Tambacounda
`KAR`	Karang	Land	Fatick
`ZIG`	Ziguinchor Airport	Air	Ziguinchor

Any other post must be added to this list (code change, To be developed) before it is connected.

The phasing below is a **Recommendation**; it follows that of the page *Deployment plan — high-availability infrastructure in Senegal*, which details the entry and exit criteria and the acceptance tests. It follows the principle of the proposal note (`docs/PROPOSITION-REMPLACEMENT-SECURIPORT.md`): a pilot at AIBD in parallel with the system in place, then a switchover in waves,

without interrupting border checks. It adds the build of the high-availability infrastructure and the application migration, which the note did not include. No schedule has been set; the durations in the plan are indicative.

Phase	Scope	Prerequisites	Exit criterion
0. Scoping and legal	Contract, CDP formalities, agreements with Sénégal Numérique (site A) and the commercial data center (site B), choice of site C, collection mandate for the security fee if the `OPERATEUR` flow is selected	"To be checked" items below handled	Agreements signed; written decision of the competent authorities
1. Build of sites A, B and C	Power, racks, network, inter-site links, HSMs, runtime platform, database, object storage, backups, monitoring and SOC	Phase 0 completed; hardware delivered	A → B failover and restoration from site C passed
2. Application migration	Move from the Netlify demonstrator to the sovereign stack (containerized API, PostgreSQL and object storage, secrets in HSM, one domain per platform), in parallel with phase 1	Priority developments started	Functional acceptance in pre-production; load test; penetration test
3. Pilot	AIBD: a few counters and eGates at arrivals and departures, in parallel with the system in place; AIBD post server; carrier gateway open to volunteer airlines; security fee reconciliation	Phases 1 and 2 validated; CDP formalities completed; certified readers and SDKs integrated; pilot officers trained	Pilot acceptance passed (see the acceptance table); no blocking incident over an agreed observation period; failover drill passed
4. Rollout in waves	Wave 1: full AIBD, Ziguinchor Airport, Autonomous Port of Dakar. Wave 2: Rosso, Kidira, Karang (VRS plate readers, scanners); decommissioning of the previous system	Pilot review; post servers installed; main and backup links validated; offline mode validated for wave 2	Acceptance per post, including a link-cut test; data takeover under the contractual clauses; final go-live report
5. Operations	24/7 monitoring, maintenance, recovery drills twice a year, quarterly restoration	Go-live report	Annual review

Hosting in Senegal

Principles

- **Sovereignty**: nominative and biometric data, as well as the keys that protect it, is hosted on national territory; no personal data is entrusted to a foreign cloud. The current Netlify hosting is not acceptable for real data (see *Security, personal data and compliance*).
- **Two active data centers** (Recommendation): web and API services run simultaneously in both sites; the database is replicated synchronously from site A to site B; each site can carry the full peak load alone.
- **A backup site (site C)**: it receives immutable encrypted copies and offline copies, without hosting production.

- **Minimal exposure**: only the traveler platform (portal, help, mobile apps page), the airline platform (`/compagnies/`) and the carrier gateway are exposed to the Internet, behind a WAF and anti-DDoS protection; the DPAF space (`/dpaf/`), kiosks and internal services remain on the private network. Each platform gets its own domain name (for example `dpaf.<domain>` , `www.<domain>` , `compagnies.<domain>` , `app.<domain>`).

Choice of sites (Recommendation, To be checked)

Site	Role	Recommended option	Points to confirm
A	Primary, active	National data center in Diamniadio (State-owned, operated by Sénégal Numérique SA), inaugurated in 2021 and presented as "Tier III type"	No Uptime Institute certification publicly confirmed; capacity, service levels, physical security and agreement **To be checked**; no commitment to date
B	Secondary, active	Tier III-certified commercial data center on a separate site and power grid: Sonatel (Orange) in Rufisque as first choice, Yas (formerly Tigo) in Diamniadio as the alternative	Rufisque certification announced but **To be checked**; Yas Diamniadio Uptime Institute Tier III certified in 2017 (validity **To be checked**) but located in the same area as site A
C	Backup vault	Secure State site or separate facility, outside the Dakar–Diamniadio–Rufisque area if possible	Site to be identified with Sénégal Numérique (**To be checked**)

The detailed comparison of the options is in the page *Deployment plan — high-availability infrastructure in Senegal*.

Service objectives (architecture targets)

Indicator	Target
Central availability	99.95% or more
Data loss within a site (RPO)	None (synchronous replication)
Data loss if a site is lost (RPO)	5 minutes at most (none while synchronous replication A → B is healthy)
Recovery time if a site is lost (RTO)	One hour at most
Checking capacity at the post	100%, including without a link to the central system
Failover tests	At least twice a year
Restoration tests	Every quarter

These values are design objectives, not contractual commitments.

Network links

Item	Recommendation
Internet for central sites	Two different carriers per site, over separate physical paths
Link between sites A and B	Dark fiber or dedicated link over two separate physical paths, encrypted, low latency (synchronous replication)

Item	Recommendation
Link to site C	Dedicated encrypted link for continuous archiving; offline copies transported under a secure procedure
Main link of posts	MPLS or encrypted VPN between each post and both central sites, automatic failover from one to the other
Backup link of posts	4G or VSAT per post, from a carrier other than that of the main link, especially for remote land posts (Kidira, Rosso, Karang)
Partners	Interpol through the NCB in Dakar (I-24/7), WAPIS, FPR, DAF and DGCPT through private links or VPN, according to the agreements (To be checked); airlines over the Internet behind the WAF
Post network	Segmented local network (counters, eGates, CCTV, administration); no control workstation directly exposed to the Internet
Outbound flows from posts	Limited to the WawID central system (HTTPS); the NFC bridge only listens on the loopback interface of each workstation
Monitoring	Device status reported to **System monitoring** and to the operator's SOC

Offline mode (To be developed)

The demonstrator has **no** offline mode: a post that loses its link can no longer record crossings. The target architecture provides at each post a **post server** (type `SRV-` in **System monitoring**), made up of two nodes, to which counters and eGates are attached, and that:

- keeps an encrypted copy of the watchlists, synchronized in under one minute;
- runs the same decision engine as the central system (`public/shared/engine.js`);
- queues crossings and resynchronizes them when the link returns;
- raises a local alert on a critical match;
- allows the post to continue checks even if both central sites are unavailable.

Until this server exists, each post must have a written **manual checking procedure**, validated by the DPAF.

Backups and recovery

The strategy follows the **3-2-1-1-0** rule: three copies, on two types of media, one of them off-site (site C), one immutable or offline, and zero errors found during verification (Recommendation).

Item	Recommendation
Database	Synchronous replication from site A to site B; continuous archiving of transaction logs (point-in-time recovery); daily full and hourly incremental backups, copied to site C
Object storage (biometrics, documents)	Replication between A and B, versioning, daily copy to site C
Immutable and offline copy	Write-once storage at site C; weekly encrypted offline copy kept in a safe, on national territory

Item	Recommendation
Audit log	Daily export of the chain; periodic anchoring of the chain head with an HSM signature (To be developed)
Keys and secrets	Kept in a pair of HSMs per central site; key backup split among several custodians
Retention	5 years shown in WawID (`retentionDays`); legal period **To be checked**; automatic purge **To be developed**
Tests	Restoration tested every quarter; restoration from site C and inter-site failover during the six-monthly drills

Failure scenarios, the failover procedure and the switchback are described in the page *Deployment plan — high-availability infrastructure in Senegal*.

Legal and regulatory prerequisites

The following checks are **To be checked** with the competent authorities. They do not presume the answers; only texts already cited in the WawID documentation are mentioned as references.

Area	Authority or contact	Check to perform
Personal data protection	Personal Data Protection Commission (CDP)	Determine the regime applicable to the processing under Law No. 2008-12 of 25 January 2008 (declaration or authorization request, in particular for biometrics and public security processing) and complete the formalities **before** any production use; set the retention period (5 years by default in WawID), the contact point for access and rectification rights, and the hosting conditions.
Border control ownership	Ministry of the Interior, Air and Border Police Directorate (DPAF)	Appointment of the data controller; validation of procedures (escalation, entry refusal, degraded mode); check whether a regulatory act creating the processing is required.
Security fee	Ministry of Economy and Finance, General Directorate of Public Accounting and Treasury (DGCPT)	Application of Decree No. 2011-652 of 26 May 2011, the interministerial order of 3 February 2012, Decree No. 2012-519 of 18 May 2012 and the decision of 4 December 2012; new allocation act if the operator share changes; flow of receipts from the Receiver General of the Treasury. The "airlines → WawID → Public Treasury" flow (WawID's default setting) differs from this framework: it requires a **new regulatory text** and a **collection-mandate agreement** between the State and WawID before any implementation.
API and PNR data	Airlines, civil aviation authority (ANACIM), DPAF	National legal basis requiring API/PNR transmission (reference: ICAO Annex 9); transmission and retention conditions; information of airlines.

Area	Authority or contact	Check to perform
Police files	Interpol NCB, national WAPIS platform, departments holding the FPR and the passport register (DAF)	Access agreements, connection conditions, clearances.
Existing contract	Contracting authorities, State Judicial Agency	Transfer, reversibility and data ownership clauses of the system currently in place (see the proposal note).
Public procurement	Contracting authorities	Procedure compliant with the Public Procurement Code or the law on public-private partnership contracts.
Telecommunications (option)	ARTP	To be consulted only if the project uses telecommunication means subject to authorization (for example private radio or satellite links).
Mobile apps	Data controller	Information notices and privacy policy required by the app stores, consistent with the CDP formalities.

Hardware per post type

Per workstation

The manufacturer references cited in the code (Regula, Thales/Gemalto, ARH, Innovatrics) are examples, not decisions. Hardware selection is a matter for the procurement procedure; biometric SDKs must be certified (NIST performance, ISO/IEC 30107-3).

Equipment	Counter	eGate / ABC kiosk	DPAF space	Post server
Computer with a recent browser (Chrome or Edge recommended)	Yes	Yes (kiosk mode)	Yes	—
Node.js 22 (for the NFC bridge)	Yes	Yes	—	Yes
Full-page document reader with RFID/NFC reader (MRZ + ICAO 9303 chip)	Yes	Yes	—	—
4-finger fingerprint scanner	Yes	Depending on model	—	—
Camera for face capture	Yes	Yes (front-facing)	—	—
Iris camera	Optional	—	—	—
QR scanner gun (e-Visa)	Recommended	—	—	—
Traveler touch screen	—	Yes	—	—
Gate controller	—	Yes	—	—

Equipment	Counter	eGate / ABC kiosk	DPAF space	Post server
UPS	Yes	Yes	Yes	Yes
Network link	Post network	Post network	Private network	Main link + backup

Per post type

The device types tracked in **System monitoring** are: counter terminals (`TRM-`), CCTV cameras (`CCTV-`), 3D baggage scanners (`LGS-` , air and sea posts), VRS plate readers (`ANPR-` , land posts) and local servers (`SRV-` , one per post).

Post type	Posts	Specific equipment	Points of attention
Air	AIBD, Ziguinchor	Counters, eGates and ABC kiosks at arrivals and departures, biometric boarding corridor, CCTV, baggage scanners	Traffic peaks; carrier gateway; security fee
Sea	Autonomous Port of Dakar	Counters at the passenger terminal, CCTV, scanners	Grouped ship arrivals; manifests uploaded as CSV if the operator is not connected
Land	Rosso, Kidira, Karang	Counters, plate readers (VRS), CCTV	Fragile network link; power cuts; degraded mode essential

Central sites

Quantities are to be sized during the detailed study; the full bill of materials is in the page *Deployment plan — high-availability infrastructure in Senegal*.

Equipment	Site A	Site B	Site C
Routers (two carriers), firewalls, switches	In pairs	In pairs	Depending on the site
Load balancers and WAF	Pair	Pair	—
Runtime platform servers (Kubernetes or equivalent)	3 control-plane nodes + worker nodes to be sized	Same	—
Database servers	At least 2	At least 2	—
Object storage	At least 4 nodes	At least 4 nodes	Immutable storage
HSM	2	2	Key backup in a safe
Backup	Backup server	Backup server	Immutable appliance, offline tapes or disks
Bastions and monitoring collectors	2	2	—

Power supply (Recommendation)

Each counter, kiosk, network device and post server is protected by a **UPS** whose autonomy is sized according to the frequency and duration of power cuts observed on site and, for isolated posts, by a generator. Running on battery must allow ongoing checks to be completed and equipment to be shut down cleanly.

Installing a counter workstation

1. Prepare the workstation

1. Install a maintained operating system, with security updates, a user session without administrator rights and automatic screen locking (Recommendation).
2. Install a recent version of Chrome or Edge.
3. Install the drivers for the document reader, fingerprint scanner and camera supplied by the manufacturer.
4. Install **Node.js 22** (required by the NFC bridge).

2. Install the NFC bridge

The `tools/nfc-bridge.mjs` bridge connects the browser to the document reader. It only listens on `127.0.0.1` (Existing).

Copy `tools/nfc-bridge.mjs` to the workstation, for example to `C:\WawID\` or `/opt/wawid/`, then set the bridge variables:

Variable	Value for Senegal
`NFC_BRIDGE_PORT`	`8765` (default), unless there is a conflict on the workstation
`NFC_BRIDGE_MODE`	`simulation` for tests; the driver name once the real reader is integrated
`NFC_BRIDGE_ORIGIN`	The exact production address of the DPAF space and kiosk, for example `https://<wawid-domain>` ; do not leave demo origins or `localhost` in operations

Then start the bridge:

```
NFC_BRIDGE_ORIGIN="https://<wawid-domain>" node tools/nfc-bridge.mjs
```

The message "Pont NFC WawID : http://127.0.0.1:8765 (mode ...)" confirms startup.

Have the bridge start automatically with the workstation (Windows service or systemd unit, Recommendation). Example systemd unit:

```
[Unit]
Description=WawID NFC bridge
After=network.target

[Service]
Environment=NFC_BRIDGE_ORIGIN=https://<wawid-domain>
ExecStart=/usr/bin/node /opt/wawid/nfc-bridge.mjs
```

```
Restart=always
User=<service-account>

[Install]
WantedBy=multi-user.target
```

> **To be developed**: in the current code, any mode other than `simulation` answers "Aucun pilote de lecteur configuré" (no reader driver configured). For a real reader, the bridge's `readFromReader()` function must be replaced by a call to the manufacturer's SDK or by a PC/SC + ICAO 9303 stack (BAC/PACE access, DG1, DG2, EF.SOD, passive authentication against CSCA certificates from the ICAO PKD). See *Deployment and configuration*.

3. Configure the workstation in WawID

1. Open `https://<wawid-domain>/dpaf/` and sign in with an administrator account (the **Settings** page is reserved for it).
2. Open **Settings → NFC reader on this workstation**.
3. Enter the **Bridge address** (`http://127.0.0.1:8765` by default), click **Save** then **Test**. The expected result is "Connected" followed by the reader name.
4. The setting belongs to the workstation's browser: redo it if the browser or the user profile changes.
5. Sign out of the administrator account; the officer then signs in with their own account.

4. Test the workstation

1. With a training account, open **New check**.
2. Read a test passport: the MRZ and check digits are displayed.
3. Click **Read chip**: the panel must **not** show "(simulated)" on an operational workstation.
4. Enable the camera and capture a face; capture fingerprints.
5. Run the checks without confirming the decision, or confirm only on an acceptance environment.

Installing an eGate kiosk

Prepare the kiosk computer like a counter workstation (browser, drivers, Node.js 22) and install the NFC bridge on the same machine (step 2 of the counter installation). Then start the browser in **kiosk mode** on the kiosk address, with the device identifier, for example:

```
chrome --kiosk "https://<wawid-domain>/gate/?gate=<DEVICE_ID>"
```

Then:

1. On the **Device activation** screen, enter the **Device key** (value of `WAWID_GATE_KEY`) and click **Activate**. The key is stored in the kiosk browser. Only an authorized technician enters it; it is never displayed or written down at the kiosk.
2. If the address does not specify the device, choose it on **Choose device**. Check the name and zone shown at the top right.

3. In the DPAF space, open **eGates & ABC** and check that the device card is **In service**, in automatic mode, with a recent heartbeat.
4. Test a remote close and reopen: the kiosk must show **Gate closed** and then return to the welcome screen within 15 seconds.
5. Set the gate parameters (face threshold, minimum age, eligibility, liveness detection, travel authorization check) according to the policy validated by the DPAF, with **Settings → Apply remotely**.

Open points for production:

- **To be developed**: the **Simulator (demo)** panel shown below the welcome screen must be hidden on operational kiosks.
- **To be developed**: there is no screen yet to declare a new eGate; devices come from the demo dataset.
- **Recommendation**: replace the single shared key `WAWID_GATE_KEY` with per-device certificates (mTLS), as planned in *Security, personal data and compliance*.

For the traveler journey and messages, see *eGate kiosk: setup and journey*.

Installing the DPAF space

The DPAF space only requires a recent browser on a workstation of the private network.

1. Provide analysts, supervisors, administrators and Treasury officers with a maintained workstation, recent browser and screen locking.
2. Give them the address `https://<wawid-domain>/dpaf/` (the former `/admin/` address still works); add it to the bookmarks. The traveler portal has no link to the DPAF space.
3. In production, restrict DPAF space access to the private network or a VPN (Recommendation); the `noindex, nofollow` tag is already applied (Existing).
4. For Treasury officers, provide access from DGCPT premises according to the chosen security policy.

Software installation of the platform

Prerequisites

- **Node.js 22**, version 22.18 or later (direct execution of `.mts` files).
- Git and access to the repository.
- For the demonstrator: a Netlify account. No database is required.

Local installation and tests

```
cd wawid
npm install
npm test          # compiles the documentation then runs the tests
npm run dev       # http://localhost:8888
```

- `npm run build` compiles the documentation (`docs/content`) into `netlify/lib/docs-content.mjs` .
- `npm test` compiles the documentation then runs `node --test` on the business engine, the API and PAXLST/PNR decoding.
- `npm run dev` starts the local server; the port is changed with `PORT` .
- Locally, data is kept in memory and development accounts and keys are created automatically (see *Deployment and configuration*). These defaults do not exist on Netlify.

Netlify deployment (demonstrator)

Setting	Value (`netlify.toml` file)
Base directory	`wawid/`
Build command	`npm run build && npm test`
Publish directory	`public`
Functions	`netlify/functions` (esbuild bundler)
Node version	`NODE_VERSION = "22"`

A failing test blocks the deployment.

Environment variables

The names below are those read by the code (`netlify/lib/auth.mts` and `netlify/lib/connectors.mts`).

Variable	Required	Purpose
`SESSION_SECRET`	Yes	Signs sessions and e-Visa QR codes (and former WawPass receipts). Without it, the API returns an error.
`WAWID_ADMIN_PASSWORD`	Yes	Password of the `admin` account, created on the first sign-in if no account exists.
`WAWID_DEMO_PASSWORD`	No	Creates the demo accounts `agent` , `analyste` , `superviseur` , `tresor` . **Never set it in production.**
`WAWID_GATE_KEY`	Yes for eGates	eGate and kiosk key (`X-Gate-Key` header).
`WAWID_CARRIER_KEYS`	Yes for the gateway	Airline keys: `IATA_CODE:key` separated by commas.
`WAWID_CONNECTOR_<ID>_URL`	No	URL of a connector (`INTERPOL_SLTD` , `INTERPOL_NOM` , `WAPIS` , `FPR` , `JUD` , `ETATCIVIL` , `PAXLST`).
`WAWID_CONNECTOR_<ID>_TOKEN`	No	`Authorization: Bearer` token of the connector.

Variables of the local server and the NFC bridge: `PORT` , `NFC_BRIDGE_PORT` , `NFC_BRIDGE_MODE` , `NFC_BRIDGE_ORIGIN` .

Setting them with the Netlify CLI (values replaced by placeholders):

```
        netlify env:set SESSION_SECRET "<RANDOM_SECRET_64_CHARACTERS>" --context
production --secret
        netlify env:set WAWID_ADMIN_PASSWORD "<ADMIN_PASSWORD>" --context production --
secret
        netlify env:set WAWID_GATE_KEY "<EGATE_KEY>" --context production --secret
        netlify env:set WAWID_CARRIER_KEYS "HC:<HC_KEY>,AF:<AF_KEY>" --context
production --secret
```

On Netlify, a secret variable must be entered **per deploy context** (at least Production) with the **Functions** scope; redeploy after every change.

Secrets handling

Generate each secret and each key randomly, for example:

```
        node -e
"console.log(require('node:crypto').randomBytes(48).toString('base64url'))"
```

Rules to apply:

1. Use a different value for each purpose (session, eGates, each airline) and for each environment (demo, acceptance, production).
2. Never put a secret in the code, the documentation, a ticket or a messaging app; send keys to airlines and technicians through a secure channel.
3. Keep a register (without the values): who holds which key, creation date, planned renewal date.
4. Changing `SESSION_SECRET` signs out every user and **invalidates every e-Visa QR code (and former WawPass receipt) already issued**: plan this operation.
5. In national production, keys are kept in an HSM (To be developed).

Initial accounts and passwords

1. Before the first deployment, set `WAWID_ADMIN_PASSWORD` (and do not set `WAWID_DEMO_PASSWORD`).
2. Sign in once with the `admin` account: it is created at that moment, if the database contains no account.
3. In **Users & roles**, immediately create a nominative account for each administrator, then the officers' accounts, with an initial password of at least 10 characters handed over in person.
4. Change the `admin` account password in **Users & roles → Edit**: changing `WAWID_ADMIN_PASSWORD` afterwards has **no effect** on an existing account. Keep this account for exceptional operations.
5. Users cannot change their own password in this version: changes go through an administrator.

Mobile apps

The **WawID e-Visa** app exists in two versions:

- `ios/`: the original SwiftUI app (iPhone), published by the "WawID iOS → TestFlight" workflow;
- `mobile/`: the new Flutter app for **iPhone and Android** (MRZ scan with the camera, NFC chip reading with BAC/PACE, reading DG1 and DG2 and checking integrity against the SOD, stay details, demo payment, tracking with QR code, French and English). It is being finalized.

Item	Value
App identifier	`sn.wawtelecom.wawid.visa` (iOS and Android)
iPhone	iPhone 7 or later, iOS 15.5 or later
Android	Android 7 (API 24) or later, with NFC
API address	`https://wawid-waw-telecom.netlify.app` by default; can be changed at build time with `--dart-define=WAWID_API=https://<wawid-domain>`

Before any public release for Senegal, the API address must point to the national platform.

Automated publishing

The GitHub Actions workflow "WawID mobile (Flutter) → APK / TestFlight" (`.github/workflows/wawid-mobile.yml`) runs Flutter analysis and tests, builds an installable **APK** and an **AAB** for Google Play, and sends the iOS version to TestFlight when the Apple secrets are present.

GitHub secret	Use
`APPLE_TEAM_ID`, `ASC_KEY_ID`, `ASC_ISSUER_ID`, `ASC_KEY_P8_BASE64`	Signing and TestFlight upload (App Store Connect API key with the App Manager role)
`ANDROID_KEYSTORE_BASE64`, `ANDROID_KEYSTORE_PASSWORD`, `ANDROID_KEY_ALIAS`, `ANDROID_KEY_PASSWORD`	Production Android signing; without them, the APK is signed with a test key (direct installation only)
`PLAY_SERVICE_ACCOUNT_JSON`	Automatic upload of the AAB to the Google Play "internal testing" track

Installing the app on iPhone

1. **Today (TestFlight)**: the tester receives a TestFlight invitation (internal testers up to 100, or an external group after Apple's "Beta App Review"). They install the **TestFlight** app from the App Store, accept the invitation, then install **WawID e-Visa**.
2. **Later (App Store)**: publication after Apple's review, with store listing, screenshots, privacy policy and justification of NFC use.
3. NFC only works on a real iPhone.

Installing the app on Android

1. **Tests (APK)**: download the APK from the workflow artifacts, copy it to the phone, allow installation from that source, then install it. Reserve this method for identified testers.
2. **Google Play internal testing**: with the keystore and service account configured, the AAB is published to the internal track; testers install it from the Google Play invitation link.

3. **Production**: publication on Google Play after validation (store listing, privacy policy, data safety form).
4. Check that **NFC is enabled** in the phone settings.

NFC tips for testers

- Passport closed, flat, without a cover; phone without a thick case; away from metal surfaces.
- iPhone: bring the top of the phone close. Android: the NFC antenna is usually on the back; its position varies by model.
- Stay still during reading; if it fails, check the MRZ (a single error prevents the chip from opening).

Developer account

Recommendation: publish under an **organization** developer account in the name of the entity that will own the app (WAW Telecom or the Senegalese administration, depending on the decision taken), not under a personal account. Enrollment conditions (company documents, D-U-N-S number for Apple, organization verification for Google) are **To be checked** with Apple and Google. Keep the production Android keystore and the App Store Connect keys in the secrets register: losing them prevents updates.

> **To be checked**: verification of the chip signature against CSCA certificates (ICAO PKD) in the Flutter app, and the exact list of data sent to the server, must be confirmed when the app is delivered.

Airline onboarding

1. **Inform** each airline serving AIBD and Ziguinchor: hand over the specification `docs/API-TRANSPORTEURS.md` and the page *API reference (airlines, devices, connectors)*.
2. **Appoint** a technical contact and a security contact per airline.
3. **Generate** one key per airline and add it to `WAWID_CARRIER_KEYS` (format `HC:<HC_KEY>,AF:<AF_KEY>`), then redeploy.
4. **Send** the key through a secure channel.
5. **Create** the airline's named accounts in **Users & roles** (role **Airline**, airline selected from the list) and send the address of the airline platform (`https://<wawid-domain>/compagnies/`) with the *Airline platform* page.
6. **Test** on the acceptance environment: the airline sends a test PAXLST and PNR, for example:

```
curl -X POST https://<wawid-domain>/api/carrier/messages \
  -H "X-Carrier-Key: <AIRLINE_KEY>" -H "Content-Type: text/plain" \
  --data-binary @paxlst.edi
```

Then:

1. **Check** the **202** response and the `MSG-…` reference, then, in **Carrier gateway**, the **ACCEPTÉ** (accepted) status and completeness.

2. **Monitor** the **Carrier compliance** tab during the first weeks (targets: 95% punctuality, 98% completeness).

The sample messages of the DPAF space (**Receive a message → PAXLST example / PNR example**) and of the airline platform (**PAXLST example** / **PNR example**) serve as a format reference. PNRGOV (EDIFACT) messages go through a converter. A compromised key is replaced without delay.

Treasury (DGCPT) onboarding

1. Designate, with the DGCPT, the authorized officers; the administrator creates their accounts with the **Treasury officer (fee)** role.
2. The administrator checks in **Revenue split → Settings** the rate per passenger, the operator share, the monthly penalty and the **Collection flow**, and enters the **IATA rate of the month** every month, in accordance with the texts in force.
3. Choose the collection flow. `OPERATEUR` flow (airlines → WawID → Treasury): open the WawID collection account, send its details to the airlines, define who validates declared payments (**Payments to validate** tab) and the schedule of transfers of the State's share to the Receiver General of the Treasury (**Transfers to the State** tab, on the 15th of month M+2). This flow only applies once a new regulatory text has been adopted and a collection-mandate agreement signed with the State. `TRESOR` flow (Decree No. 2012-519): define the flow of receipts from the Receiver General of the Treasury (who enters them, within what time, with which reference).
4. Train officers on the monthly calendar (collection from the 1st to the 9th, invoices on the 10th, due date at the end of the following month), on validating declared payments and on **Reconciliation**.
5. Validate the CSV export format (`wawid-redevance.csv`) with public accounting.

> **To be developed**: in the demonstrator, invoices come from the demo dataset. Issuing real invoices (airline declarations, IATA rate, WawID count) must be implemented before operations.

Training plan (Recommendation)

Audience	Content	Material
DPAF officers (counter)	Sign-in, full check, chip reading, biometrics, justified decisions, QR, common situations, degraded mode	*User guide by profile*; demo scenarios
Supervisors	Escalations, eGates, dashboard, cases, audit log	*eGates and ABC kiosks*; *Dashboard*
Analysts	Threat Monitor, lists, rules, APIS/PNR, e-Visa, identity networks	*Threat Monitor, cases, watchlists, rules and connectors*; *APIS and PNR*
Administrators	Accounts, settings, NFC bridge, keys, backups, incidents, failover and restoration	This guide; *Deployment and configuration*; *Deployment plan — high-availability infrastructure in Senegal*

Audience	Content	Material
Post technicians	Installation of workstations and kiosks, first-level diagnosis	This guide; *eGate kiosk: setup and journey*
Treasury officers	Calendar, payment validation, collections, transfers to the State, reconciliation, export	*Security fee*
Airlines	Airline platform, formats, schedule, responses, compliance, security fee payment declaration	*Airline platform*; `docs/API-TRANSPORTEURS.md`

Run training on a demonstration environment separate from production, and appoint in each post a **WawID focal point** trained in first-level diagnosis.

Support and escalation (Recommendation)

Level	Actor	Scope
L1	Post WawID focal point	Workstations, readers, cameras, kiosks, usage procedures
L2	Operator support and 24/7 network operations center (NOC)	Accounts, settings, network, NFC bridge, connectors, application incidents, status of central sites and links
L3	Development team	Software defects, fixes
Security	Security officer and SOC	Security incident, key leak, broken audit chain
Crisis	Crisis unit (operator, DPAF)	Loss of a central site, ransomware, prolonged degraded mode

Contact details, hours and response times for each level are set with the DPAF before go-live and displayed in each post. Planned maintenance is announced in advance to posts, airlines and the Treasury.

Acceptance testing

No.	Test	Expected result
1	`npm test` on the delivered version	All tests pass
2	Officer sign-in, sign-out, expired session	Access matches the role; actions written to the audit log
3	Counter check of a real biometric test passport	Valid MRZ; chip read without "(simulated)"; passive authentication valid
4	Check of an altered MRZ (**Forged MRZ** scenario)	Chip and MRZ different; recommendation of secondary inspection or higher
5	Watchlist match	Alert in the **Threat Monitor**; supervisor escalation
6	eGate crossing by an eligible traveler	Gates open; crossing recorded in the **eGate / ABC** channel

No.	Test	Expected result
7	eGate crossing by a minor	Referral to the counter with the expected reason
8	Remote closing of an eGate	**Gate closed** shown within 15 seconds
9	e-Visa application on the portal, information request, reply, approval	Correct successive statuses; signed QR verified at the counter
10	e-Visa application from the iPhone and Android app	Chip read; application visible in the DPAF space with the **Chip (NFC)** field
11	**Security fee** card on the traveler portal	Information page "You have nothing to pay", fee included in the ticket; no form and no payment request; `POST /api/public/pass` answers `410`
12	PAXLST and PNR message from an airline with its key	202 with no pre-screening result in the acknowledgment; API + PNR merge; flagged passengers visible in the DPAF space
13	Message with another airline's key	Rejection (422)
14	Collection recorded on an invoice by a Treasury officer	**PARTIELLE** (partial) or **PAYÉE** (paid) status; `REDEVANCE_ENCAISSEMENT` entry
15	**Audit log** check	**Chain intact**
16	Sign-in of an airline account on `/dpaf/`, then on `/compagnies/`	Refused on the DPAF space; on the platform, only its own airline's data, with no flagged passenger or score
17	Payment declared by an airline, validated by a Treasury officer, then transfer of the State's share (`OPERATEUR` flow)	Payment **Awaiting validation** then **Validated**; invoice **PARTIELLE** (partial) or **PAYÉE** (paid); `REDEVANCE_PAIEMENT_DECLARE`, `REDEVANCE_PAIEMENT_CONFIRME`, `REDEVANCE_REVERSEMENT_ETAT` entries
18	Link cut at a post (after offline mode is developed)	Checks continue; resynchronization when the link returns, with no loss or duplicate
19	Power cut at a counter	Switch to UPS; no crossing lost
20	Failover from site A to site B, then back	RTO of one hour at most; RPO met; **Chain intact** in the **Audit log** after failover
21	Full restoration from a site C copy in pre-production	Restoration successful; duration and data loss measured
22	Simulated HSM failure	No interruption of signing
23	Load test	AIBD peak of 15,000 travelers per hour carried by a single central site
24	Penetration test	No open critical or high vulnerability

Go-live checklist

1. CDP formalities completed and written decisions of the competent authorities obtained.
2. Sites A and B operational and agreements signed; site C receiving immutable and offline copies; A → B failover, switchback and restoration from site C passed.
3. 24/7 monitoring center, SOC, on-call duty and crisis unit in place; failover and restoration procedures written.

4. Demo dataset disabled; no fictitious data in the production database.
5. `SESSION_SECRET`, `WAWID_ADMIN_PASSWORD`, `WAWID_GATE_KEY` and `WAWID_CARRIER_KEYS` set; `WAWID_DEMO_PASSWORD` absent.
6. Nominative accounts created; `admin` account password changed; role review done.
7. NFC bridges installed with `NFC_BRIDGE_ORIGIN` limited to the production domain; **Connected** test on each workstation.
8. Readers, biometric SDKs and kiosks integrated and tested; simulator hidden on kiosks.
9. Connectors (Interpol, WAPIS, FPR, DAF) connected or, failing that, local copy of the lists up to date and procedure validated.
10. Airlines connected and tested, airline platform accounts created; security fee collection flow settled with the matching legal basis; Treasury officers trained.
11. Mobile apps pointing to the national API; store listings validated.
12. Degraded-mode procedure displayed in each post; post servers, backup links, UPS units and generators tested.
13. Officers trained; focal points appointed; escalation chain shared.
14. Acceptance signed; go-live report drawn up.